A Multi-Agent Large Language Model Framework for Explainable Stock Research and Investment Candidate Selection

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***Abstract*—Equity research requires analysts to combine market movements, company fundamentals, news, and risk factors before forming an investment view. These activities are time-consuming and may be inconsistent when information is collected manually from multiple sources. This paper presents the design of a multi-agent stock research assistant that coordinates specialized large language model (LLM) agents to discover candidate companies, gather financial and news evidence, evaluate risks, and produce a structured investment research report. The proposed workflow uses an orchestration agent to delegate tasks to a trending-company discovery agent, a financial researcher, and a stock-selection agent. External search and financial-data tools provide evidence for analysis, while a report-generation stage converts the final assessment into a portable PDF report and supports email delivery. The system is designed to produce a reasoned Buy, Hold, or Avoid-style assessment with supporting evidence, uncertainty, and risk disclosures rather than execute trades. This paper describes the architecture, workflow, data handling, and an evaluation plan. Since a controlled backtest and quantitative user study have not yet been documented, no claims of predictive accuracy or investment returns are made. The framework is intended as a decision-support prototype and not as financial advice.**

***Index Terms*—multi-agent systems, large language models, financial research, stock screening, explainable AI, decision support, CrewAI**

I. Introduction

Investment research involves more than observing a stock's recent price. A defensible assessment may require price and volume history, revenue and earnings trends, profitability, valuation, leverage, cash flow, company announcements, market news, and downside risks. Gathering these signals from different sources can be labor-intensive, particularly when an analyst must repeat the process across several companies.

Large language models (LLMs) can interpret unstructured text and use tools to retrieve external evidence, but a single unconstrained model may mix responsibilities, overlook contradictory information, or generate unsupported claims. The ReAct paradigm combines language-model reasoning with actions that obtain observations from external tools, providing a useful conceptual foundation for tool-using agents [1]. A multi-agent design further separates tasks into narrower roles and makes intermediate outputs easier to inspect.

This paper describes a Stock Picker Agent: a multi-agent investment-research workflow built around company discovery, financial research, candidate selection, report generation, and email delivery. The system is designed to help a user organize evidence and compare candidate companies. It does not guarantee future price movement, and its output must not be interpreted as a substitute for professional financial advice.

The contributions of this work are:

- A modular workflow that separates candidate discovery, financial research, and final stock assessment.
- An evidence-oriented report structure that records supporting signals, risks, and unavailable data.
- A practical end-to-end design that connects research outputs to PDF generation and email delivery.
- An evaluation protocol for testing factual grounding, completeness, reproducibility, latency, and the quality of recommendations without inventing performance results.

II. Related Work

A. Tool-Using Language Models

Yao *et al.* introduced ReAct, which interleaves reasoning and external actions so that a language model can update its task state using observations from tools [1]. For financial research, this pattern is relevant because company information is distributed across news pages, financial statements, and market-data services. Tool access, however, does not itself ensure correctness: retrieved material can be stale, incomplete, or contradictory, so outputs still require validation.

B. Multi-Agent Task Decomposition

Multi-agent LLM workflows assign distinct responsibilities to specialized agents and coordinate their outputs. In the proposed system, discovery, financial analysis, and final selection are separated to reduce the burden placed on a single prompt and to make each stage independently inspectable. The design uses CrewAI as the agent-orchestration framework; the framework's role here is operational coordination, not proof that the resulting recommendations are profitable.

C. Financial Decision Support

Classical portfolio theory emphasizes the relationship between risk and return rather than treating expected return in isolation [2]. The proposed prototype is narrower than a portfolio optimizer: it produces a research assessment for

candidate stocks and does not currently claim to estimate an optimal portfolio allocation. Its risk-analysis stage is included to ensure that positive narratives are considered alongside adverse signals.

D. Research Gap and Motivation

Many lightweight stock-research prototypes focus on a single LLM response or a summary of recent news. The proposed workflow instead separates discovery, financial investigation, risk review, and final synthesis, then packages the result into a report. This is an architectural contribution; its practical value must be established through systematic evaluation rather than assumed from the use of multiple agents.

III. System Overview

The system accepts a research request, such as a market or sector focus, or a company ticker when the workflow is configured for a specific company. A manager/orchestrator controls task order and passes intermediate results to the relevant agents. The core conceptual workflow is shown in Fig. 1.

A. Manager / Orchestrator

The manager receives the research objective, delegates subtasks, and assembles intermediate results. It is responsible for maintaining a consistent company identity across stages and for ensuring that the final report is based on the outputs of the research agents rather than on an unrelated free-form answer.

B. Trending Company Discovery Agent

This agent identifies candidate companies using available web-search results and market-related information. Its output should include the company name, ticker, reason for inclusion, source links, and retrieval date where available. A company being frequently discussed or trending is treated as a discovery signal, not as evidence that it is undervalued or suitable for investment.

C. Financial Research Agent

The financial researcher collects available market and fundamental information. Relevant fields may include recent price performance, revenue and earnings trends, profitability, valuation indicators, debt, and cash flow. A web-search tool such as Serper and financial-data sources such as Yahoo Finance may be integrated where configured. Data availability varies by company, exchange, source, and API access; missing values must be marked as unavailable rather than inferred.

D. Stock Selection and Risk Review

The selection stage compares the available evidence and creates a structured assessment. It considers positive signals, negative signals, and uncertainty. The output may use labels such as *Buy*, *Hold*, or *Avoid*, but these are heuristic research labels and must not be presented as calibrated probabilities or guaranteed outcomes unless validated against a defined dataset. The agent should state the reasons behind its assessment and flag important missing or conflicting information.

E. Report Generation and Delivery

After synthesis, the system formats the result into a PDF report and can deliver it through an email step. A report should identify the company and ticker, timestamp the research, summarize the thesis, list supporting evidence and risks, cite sources, and include a disclaimer. Email delivery is a presentation and workflow feature; it does not independently validate the recommendation.

IV. Methodology

A. Workflow

The proposed process consists of the following stages:

1) **Input normalization:** parse the user's market, sector, or ticker request and establish the scope of analysis.
2) **Candidate discovery:** retrieve candidate companies and record the reason each candidate entered the research set.
3) **Evidence collection:** retrieve market, fundamental, and news information using configured external tools.
4) **Financial analysis:** summarize the available financial indicators and their direction over the relevant time window.
5) **Risk review:** identify adverse news, business risks, valuation concerns, leverage, and evidence gaps.
6) **Final synthesis:** compare signals, produce a clearly qualified assessment, and preserve supporting evidence.
7) **Report delivery:** generate a PDF and, if enabled, email the report to the intended recipient.

B. Evidence Representation

To improve traceability, each material claim in the final report should be represented with the following fields:

$$e_i = (c_i, s_i, t_i, d_i, q_i), \tag{1}$$

where c_i is the claim or observation, s_i is its source, t_i is the retrieval timestamp, d_i is the relevant date of the underlying data, and q_i is a quality or availability note. This representation is a design recommendation for the system and should be implemented in the data structures if not already present.

The final report should distinguish between (i) source-derived facts, (ii) the agent's interpretation of those facts, and (iii) missing information. When sources disagree, the disagreement should be exposed rather than silently resolved. A source URL and date should accompany news-derived claims wherever possible.

C. Candidate Assessment

A transparent scoring rubric can help organize evidence, but weights must be selected and validated rather than invented after observing outcomes. As an initial design, the system may report separate qualitative dimensions:

- **Fundamental condition:** direction and consistency of reported financial indicators.
- **Market behavior:** recent price and volume context over a declared period.
- **News and catalysts:** relevant positive and negative developments with source dates.

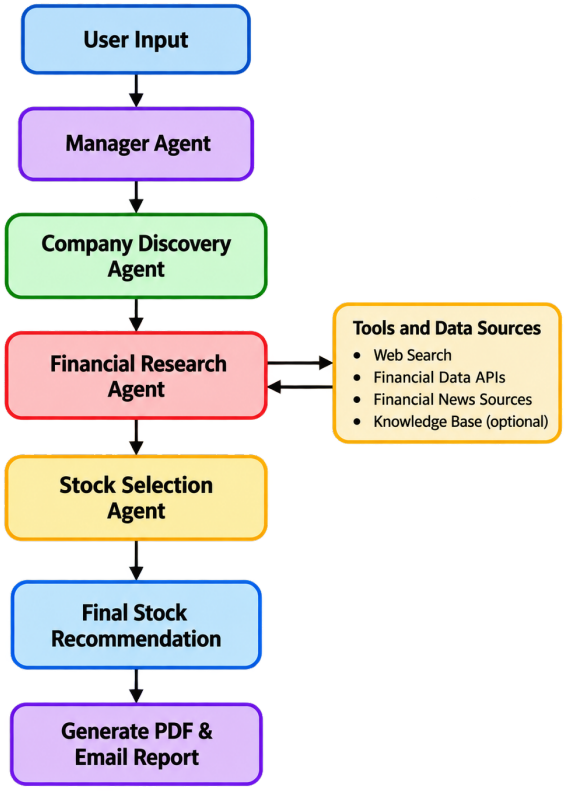


Fig. 1. Workflow of the proposed multi-agent Stock Picker Agent.

- **Risk and uncertainty:** valuation concerns, debt, business exposure, conflicting reports, and missing data.

These dimensions should remain separately visible in the report. If a single aggregate score is added, its formula, scale, weights, and validation procedure must be documented. No numerical scoring formula or learned predictive model is claimed in this paper because verified implementation details and evaluation data were not supplied.

D. Reliability Controls

The following controls are recommended for implementation and evaluation:

- Validate ticker and company identity before combining data from different sources.
- Include retrieval timestamps and avoid describing old data as current.
- Require a source for material factual claims and mark unsupported claims as unverified.
- Preserve unavailable values as missing rather than filling them with model-generated guesses.
- Apply a structured output schema and check required report fields before PDF generation.
- Keep the final decision advisory and require human review before any financial action.

V. Implementation Considerations

The project context indicates the use of CrewAI for multi-agent orchestration, Groq-hosted LLMs, Serper for web search, financial research tools, PDF generation, and email delivery. Exact model identifiers, prompt configurations, API versions, and data-provider settings should be documented from the actual repository before final submission. They are intentionally not fixed here because such details can change between runs.

A practical implementation can represent each stage as an agent/task pair with explicit inputs and outputs. Intermediate outputs should be stored in a shared state object or structured task result. The final synthesis stage should consume the discovery and research outputs, not repeat the entire research process without reference to previous results. Failures from unavailable APIs, rate limits, and malformed responses should be surfaced in the report or logs.

VI. Evaluation Plan

A research paper should evaluate both the quality of the generated analysis and the reliability of the underlying workflow. Since no benchmark results, completed backtest, or measured user-study results were supplied for this draft, Table I defines an evaluation plan rather than reporting experimental findings.

TABLE I
PROPOSED EVALUATION PROTOCOL

Dimension	Measurement approach
Factual grounding	Manually audit sampled claims against cited source material; report supported, contradicted, and unverifiable claim rates.
Completeness	Check whether required market, fundamental, news, risk, timestamp, and disclaimer fields are present.
Source traceability	Measure the proportion of material claims with working source links and retrieval dates.
Reproducibility	Repeat identical requests and compare retrieved evidence, intermediate outputs, and final labels.
Operational performance	Record end-to-end runtime, API failures, retry count, and cost per completed report.
Recommendation quality	Use a time-separated historical evaluation with a predeclared universe, benchmark, transaction costs, and no look-ahead leakage.
Usability	Ask reviewers to rate clarity, usefulness, and ability to trace the rationale to evidence.

A. Recommended Historical Evaluation

If the system is intended to make claims about investment performance, a backtest must be conducted before such claims are included. The experiment should define the stock universe and selection dates in advance, use only information available at each decision time, specify a holding period and rebalancing rule, and include transaction costs and a benchmark. Appropriate metrics may include cumulative return, annualized return, volatility, maximum drawdown, and risk-adjusted performance. Results should be compared with a simple baseline, such as an equal-weight selection or a predefined market benchmark. Survivorship bias and look-ahead bias must be considered.

VII. DISCUSSION

The principal benefit of the design is task decomposition. Discovery and financial research have different objectives, while risk review provides a dedicated opportunity to challenge an overly positive narrative. The final report can make the process more usable by packaging the rationale and its evidence into a single artifact.

The architecture also introduces limitations. Agent outputs depend on the quality, recency, and coverage of external data. An LLM may misinterpret financial terminology, summarize a source incorrectly, or produce a plausible but unsupported explanation. Multiple agents can repeat the same underlying error, so agent count alone should not be treated as a reliability guarantee. External APIs may change, impose rate limits, or omit relevant companies and metrics. Finally, a readable report is not proof of predictive skill.

For these reasons, the system should be positioned as a research assistant that helps structure and summarize information. It should not autonomously execute trades, imply certainty, or claim superior investment returns without controlled evidence.

VIII. LIMITATIONS AND FUTURE WORK

This draft describes the known project concept and a proposed technical methodology; it does not report a completed quantitative experiment. The current evidence available for this manuscript does not establish accuracy, profitability, latency, cost, or user-satisfaction measurements. These must be measured on the actual implementation.

Future work includes: (1) adding a versioned and reproducible data-collection layer; (2) enforcing source-linked structured outputs; (3) developing a documented scoring rubric; (4) evaluating on historical data with strict temporal separation; (5) comparing the multi-agent workflow against a single-agent baseline; and (6) conducting a human review of factual support and report usefulness. Any extension to portfolio allocation should separately model diversification, risk tolerance, and constraints.

IX. CONCLUSION

This paper presented a multi-agent LLM framework for automating parts of stock research and candidate selection. The proposed system coordinates company discovery, financial analysis, risk review, final synthesis, PDF report generation, and email delivery. Its central design principle is to separate task responsibilities while preserving evidence, timestamps, uncertainty, and risks in the final output. The framework is a decision-support prototype, not a validated prediction engine. Future empirical evaluation is required to establish whether the architecture improves factual reliability, research efficiency, or investment outcomes.

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REFERENCES

[1] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing Reasoning and Acting in Language Models," in *Proc. International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://arxiv.org/abs/2210.03629

[2] H. Markowitz, "Portfolio Selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, Mar. 1952, doi: 10.2307/2975974.

[3] CrewAI, "CrewAI Documentation." [Online]. Available: https://docs.crewai.com/. Accessed: Oct. 30, 2026.

[4] Ran Aroussi, "yfinance: Download market data from Yahoo! Finance's API," project documentation. [Online]. Available: https://github.com/ranaroussi/yfinance. Accessed: Oct. 30, 2026.

[5] Serper, "Google Search API." [Online]. Available: https://serper.dev/. Accessed: Oct. 30, 2026.